

KodetoCareer — Technical Requirements Document (TRD)

Version 2.0 | Updated — 100% Free Infrastructure Stack

INFRASTRUCTURE CHANGE NOTICE

This version replaces all paid AWS services with permanently free alternatives. The free stack is production-grade and sufficient for 0 to 10,000 students without incurring any infrastructure cost.

OLD (Paid AWS) | NEW (Free Alternative) | Free Limit

AWS EC2 | Oracle Cloud Always Free (ARM) | 4 OCPUs, 24GB RAM — permanent

AWS RDS PostgreSQL | Neon.tech | 0.5GB DB, serverless — permanent

AWS ElastiCache Redis | Upstash Redis | 10,000 cmd/day, 256MB — permanent

AWS S3 + CloudFront | Cloudflare R2 + CDN | 10GB storage, zero egress — permanent

AWS MediaConvert | Removed (direct MP4 on R2) | N/A

AWS SES | Resend | 3,000 emails/month — permanent

MSG91 / Twilio SMS | Email OTP only | Free

AWS Certificate Manager | Cloudflare SSL | Free

AWS Secrets Manager | System env vars | Free

1. System Architecture Overview

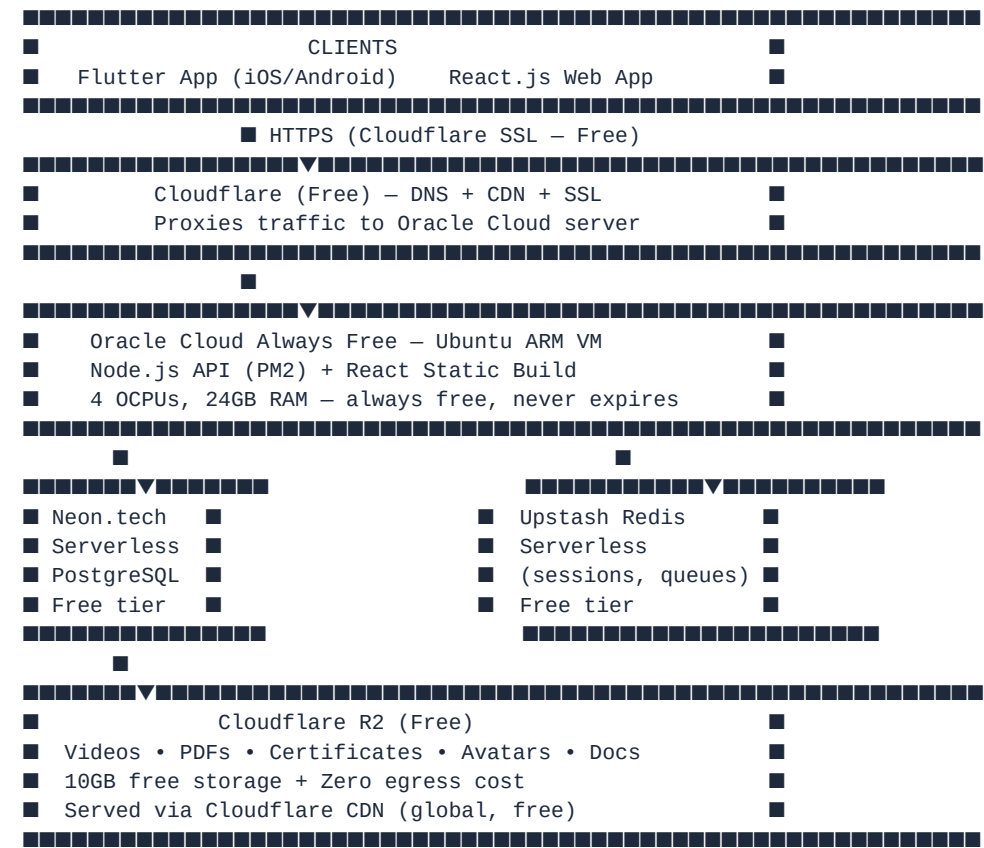
The KodetoCareer platform follows a **modular monolith** architecture deployed entirely on free-tier infrastructure that never expires.

Architecture Pattern

- **Backend:** Modular Monolith (Node.js / Express.js)
- **Frontend Web:** React.js (Single Page Application)
- **Mobile:** Flutter (iOS + Android, single codebase)
- **Database:** PostgreSQL via Neon.tech (serverless, free)
- **Cache:** Redis via Upstash (serverless, free)
- **Storage:** Cloudflare R2 (S3-compatible, zero egress fees)
- **CDN:** Cloudflare (free, global)
- **Authentication:** JWT (RS256) + Refresh tokens
- **Real-time:** Socket.io

- **Video:** Direct MP4 on Cloudflare R2 (served via CDN) + YouTube/Vimeo embed support
- **Email:** Resend (free tier, 3,000/month)

2. Infrastructure Architecture



3. Technology Stack — Detailed

3.1 Backend (API Server)

Component	Technology	Version	Rationale
Runtime	Node.js	20 LTS	Team familiarity, async I/O
Framework	Express.js	4.x	Minimal, flexible
ORM	Prisma	5.x	Works perfectly with Neon.tech PostgreSQL
Validation	Zod	3.x	Runtime schema validation
Authentication	jsonwebtoken	9.x	JWT RS256
File Upload	Multer + @aws-sdk/client-s3	Latest	R2 is S3-compatible, same SDK
Email	Resend SDK	Latest	Free 3,000/month, excellent deliverability

Queue | Bull (Upstash Redis) | Latest | Background jobs via @upstash/redis

Logging | Winston | Latest | Structured logging to file + console

API Docs | Swagger/OpenAPI | 3.0 | Auto-generated

Testing | Jest + Supertest | Latest | Unit and integration tests

Process Manager | PM2 | Latest | Production process management

3.2 Frontend Web

Component | Technology | Version

Framework | React.js | 18.x

State Management | Zustand | 4.x

Routing | React Router | 6.x

UI Components | shadcn/ui | Latest

Styling | Tailwind CSS | 3.x

Data Fetching | React Query (TanStack) | 5.x

HTTP Client | Axios | 1.x

Forms | React Hook Form + Zod | Latest

Charts | Recharts | 2.x

Rich Text | TipTap | 2.x

Video Player | Video.js | 8.x

Table | TanStack Table | 8.x

Build | Vite | 5.x

3.3 Mobile (Flutter)

Component | Technology

Framework | Flutter 3.x

State Management | Riverpod 2.x

HTTP | Dio 5.x

Local Storage | flutter_secure_storage

Push Notifications | firebase_messaging (FCM — free)

Video Player | video_player + chewie

PDF Viewer | flutter_pdfview

Offline Support | sqflite

Analytics | Firebase Analytics (free)

Crash Reporting | Firebase Crashlytics (free)

[Deep Linking](#) | [go_router](#)

3.4 Database — Neon.tech (Free)

[Component](#) | [Details](#)

[Provider](#) | Neon.tech

[Engine](#) | PostgreSQL 15

[Free Tier](#) | 0.5GB storage, 1 compute unit, unlimited projects

[Connection](#) | Neon serverless driver OR standard pg connection

[Pooling](#) | Built-in Neon connection pooling (pgBouncer compatible)

[Backups](#) | Neon automatic point-in-time recovery (7 days on free tier)

[Scale](#) | Scales to zero when idle (no cost), instant wake

Prisma connection string for Neon:

```
DATABASE_URL="postgresql://user:pass@ep-xxx.ap-southeast-1.aws.neon.tech/kodetocareer..."
```

3.5 Cache — Upstash Redis (Free)

[Component](#) | [Details](#)

[Provider](#) | Upstash

[Free Tier](#) | 10,000 commands/day, 256MB, 1 database

[Use Cases](#) | Sessions, rate limiting, Bull queues, API cache

[SDK](#) | @upstash/redis (REST-based, works without persistent connection)

[Bull Queues](#) | Use bull with ioredis connecting to Upstash Redis

Note: 10,000 commands/day is sufficient for ~500–1,000 active daily users. Upgrade to Upstash Pay-as-you-go (\$0.20 per 100K commands) only when you scale beyond that.

3.6 Storage — Cloudflare R2 (Free)

[Component](#) | [Details](#)

[Provider](#) | Cloudflare R2

[Free Tier](#) | 10GB storage, 1M Class A ops/month, 10M Class B ops/month

[Egress Cost](#) | ZERO (Cloudflare charges no egress — unlike AWS S3)

[API](#) | S3-compatible — use AWS SDK v3 with R2 endpoint

[CDN](#) | Served via Cloudflare CDN for free (global edge)

[Custom Domain](#) | e.g., cdn.kodetocareer.com → your R2 bucket

R2 connection config (S3-compatible):

```
import { S3Client } from '@aws-sdk/client-s3';
export const r2Client = new S3Client({
```

```

region: 'auto',
endpoint: `https://${process.env.CF_ACCOUNT_ID}.r2.cloudflarestorage.com`,
credentials: {
  accessKeyId: process.env.R2_ACCESS_KEY_ID!,
  secretAccessKey: process.env.R2_SECRET_ACCESS_KEY!,
},
});
// Bucket name: process.env.R2_BUCKET_NAME

```

4. Video Architecture (No Transcoding — Free)

Approach for MVP

AWS MediaConvert (paid) is removed. Instead:

Option A — Direct MP4 Upload (for admin-uploaded content):

1. Admin uploads MP4 via presigned R2 URL
2. File stored on Cloudflare R2 (up to 10GB free)
3. Served directly from Cloudflare CDN — fast delivery globally
4. No transcoding — one quality level (original upload quality)
5. Video player streams MP4 directly from CDN URL

Option B — Embed URL (YouTube/Vimeo — recommended for large videos):

1. Trainer uploads video to YouTube (unlisted) or Vimeo
2. Trainer pastes embed URL in admin panel
3. Platform embeds the video in an iframe player
4. Zero storage cost, unlimited size, CDN handled by YouTube/Vimeo

Recommended default: YouTube Unlisted for large course videos. R2 direct upload for short demo/intro clips under 100MB.

What Changes from Original TRD

- No AWS MediaConvert job creation
- No Lambda functions
- No HLS transcoding pipeline
- No adaptive bitrate (480p/720p/1080p)
- Video player uses standard MP4 or YouTube/Vimeo iframe — both work on all devices

Storage Structure on Cloudflare R2

```

kodetocareer-bucket/
├── videos/
│   ├── {courseId}/{lessonId}/{filename}.mp4  (direct uploads, max 2GB)
│   └── notes/
│       ├── {courseId}/{lessonId}/{filename}.pdf
│       └── certificates/
│           └── {year}/{studentId}/{certificateId}.pdf

```

```
■■■ documents/
■   ■■■ {studentId}/offers/{filename}.pdf
■■■ avatars/
■   ■■■ {userId}/{filename}.jpg
■■■ thumbnails/
    ■■■ {courseId}/{filename}.jpg
```

5. API Design (Unchanged from v1)

All API standards, endpoint patterns, response envelopes, and authentication remain identical to TRD v1. No changes needed here.

6. Authentication & Security (Unchanged)

All JWT, refresh token rotation, RBAC, and security requirements remain identical. No paid services involved.

7. Email — Resend (Free)

Resend Free Tier: 3,000 emails/month, 100 emails/day — permanently free.

```
import { Resend } from 'resend';

const resend = new Resend(process.env.RESEND_API_KEY);

export const sendEmail = async ({
  to, subject, html
}): { to: string; subject: string; html: string } => {
  return resend.emails.send({
    from: 'KodetoCareer <noreply@kodetocareer.com>',
    to,
    subject,
    html,
  });
};
```

Note on OTP delivery: SMS OTP (MSG91/Twilio) is removed. All OTPs are delivered via email only. This is free and sufficient for the target user base. WhatsApp OTP can be added in Version 2 via Interakt (free tier available).

8. SMS — Removed from MVP

SMS notifications (MSG91/Twilio) are removed from the MVP.

Replacement: All OTP and critical alerts go via email.

Rationale: Students have email access. Email OTP is reliable. SMS adds ■800–■2,000/month cost with no proportional benefit for MVP.

Version 2 plan: Add Twilio/MSG91 SMS only when platform is generating revenue.

9. Performance Requirements (Unchanged)

All performance targets from TRD v1 remain identical. Oracle Cloud ARM instances (4 OCPUs, 24GB RAM) comfortably handle 10,000 concurrent users — more capable than t3.medium EC2.

Caching Strategy (Updated for Upstash)

All Redis caching logic remains identical. The only change is the connection client:

```
// Use @upstash/redis for REST-based connection (works in serverless environments)
import { Redis } from '@upstash/redis';
const redis = new Redis({
  url: process.env.UPSTASH_REDIS_REST_URL!,
  token: process.env.UPSTASH_REDIS_REST_TOKEN!,
});

// Or use ioredis with Upstash Redis endpoint (for Bull queue compatibility)
import Redis from 'ioredis';
const redisClient = new Redis(process.env.UPSTASH_REDIS_URL!);
```

10. Background Job Queue (Updated)

Bull queues work with Upstash Redis via ioredis connection. All job types remain the same:

Queue | Jobs | Change

video-processing | Now only: update DB status on upload complete | MediaConvert removed

certificate-generation | PDF generation, R2 upload | S3 → R2 (same SDK)

email-notifications | Via Resend SDK | SES → Resend

sms-notifications | REMOVED from MVP | Use email instead

report-generation | Excel/PDF export | No change

11. Environment Configuration (Updated)

```
# App
NODE_ENV=production
PORT=3000
API_VERSION=v1
FRONTEND_URL=https://app.kodetocareer.com

# Database — Neon.tech (FREE)
DATABASE_URL=postgresql://user:pass@ep-xxx.neon.tech/kodetocareer?sslmode=require

# Redis — Upstash (FREE)
UPSTASH_REDIS_REST_URL=https://xxx.upstash.io
UPSTASH_REDIS_REST_TOKEN=xxx
UPSTASH_REDIS_URL=redis://xxx:xxx@xxx.upstash.io:6380 # for ioredis/Bull

# Auth (no change — self-generated keys, free)
```

```

JWT_PRIVATE_KEY=<RS256 private key – generate with openssl>
JWT_PUBLIC_KEY=<RS256 public key>
REFRESH_TOKEN_SECRET=<64 char random>

# Cloudflare R2 (FREE – S3-compatible)
CF_ACCOUNT_ID=<cloudflare account id>
R2_ACCESS_KEY_ID=<r2 access key>
R2_SECRET_ACCESS_KEY=<r2 secret key>
R2_BUCKET_NAME=kodetocareer-production
R2_PUBLIC_URL=https://cdn.kodetocareer.com # Your Cloudflare CDN custom domain

# Email – Resend (FREE 3,000/month)
RESEND_API_KEY=re_XXXXXXXXXXXX
FROM_EMAIL=noreply@kodetocareer.com

# Push Notifications – Firebase FCM (FREE)
FCM_SERVER_KEY=<firebase server key>
FIREBASE_PROJECT_ID=<project id>

# Monitoring – Sentry (FREE tier)
SENTRY_DSN=<dsn>

# App Domain
API_DOMAIN=api.kodetocareer.com

```

Removed (all were paid):

- AWS_REGION, AWS_ACCESS_KEY_ID, AWS_SECRET_ACCESS_KEY
- SES_REGION
- MSG91_API_KEY, MSG91_SENDER_ID
- MEDIA CONVERT_ENDPOINT, MEDIA CONVERT_ROLE_ARN

12. Deployment Architecture (Updated)

Production Deployment — Oracle Cloud Always Free

Oracle Cloud Free Tier (No credit card charges – permanent):

Compute:

- VM.Standard.A1.Flex (ARM Ampere)
 - 4 OCPUs
 - 24 GB RAM
 - 200 GB Boot Volume (free)
 - Ubuntu 22.04 LTS
- Services running on this VM:
 - Node.js API (PM2, 3 instances using cluster mode)
 - Nginx (reverse proxy, load balance PM2 instances)
 - React build (served as static files via Nginx)
 - Bull workers (background jobs)

DNS + CDN + SSL:

- Cloudflare (free)
 - DNS management
 - SSL/TLS (full strict)
 - CDN for static React files

■■■ DDoS protection (free)

Database: Neon.tech (external, free, serverless)
 Cache: Upstash Redis (external, free, serverless)
 Storage: Cloudflare R2 (external, free)

Oracle Cloud Setup Steps

1. Create Oracle Cloud account (no credit card required for Always Free)
2. Provision VM.Standard.A1.Flex: 4 OCPU, 24GB RAM, Ubuntu 22.04
3. Configure Security Group: open ports 80, 443, 3000
4. Install: Node.js 20, Nginx, PM2, Git
5. Configure Nginx as reverse proxy to Node.js port 3000
6. Set up Cloudflare: point domain to Oracle Cloud public IP, enable proxy
7. SSL handled automatically by Cloudflare

PM2 Ecosystem File

```
// ecosystem.config.js
module.exports = {
  apps: [{
    name: 'kodetocareer-api',
    script: 'dist/index.js',
    instances: 3, // Use 3 of 4 OCPUs for API
    exec_mode: 'cluster',
    max_memory_restart: '4G',
    env_production: {
      NODE_ENV: 'production',
      PORT: 3000,
    }
  }]
};
```

CI/CD (GitHub Actions — Free)

```
# .github/workflows/deploy.yml
name: Deploy to Oracle Cloud
on:
  push:
    branches: [main]
jobs:
  deploy:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-node@v4
        with: { node-version: '20' }
      - run: npm ci && npm run build && npm test
      - name: Deploy via SSH
        uses: appleboy/ssh-action@v1
        with:
          host: ${ secrets.ORACLE_HOST }
          username: ubuntu
          key: ${ secrets.ORACLE_SSH_KEY }
          script: |
```

```
cd /app/kodetocareer-api
git pull origin main
npm ci --production
npm run build
npx prisma migrate deploy
pm2 reload ecosystem.config.js --env production
```

13. Monitoring (Free Tools Only)

Tool | Purpose | Free Tier

Sentry | Application error tracking | 5,000 errors/month

Firebase Analytics | Mobile user behaviour | Free

Firebase Crashlytics | Crash reporting | Free

UptimeRobot | External uptime monitoring | 50 monitors, 5-min checks

PM2 Monit | Process monitoring on server | Free (built-in)

Neon Dashboard | DB query performance | Free

Cloudflare Analytics | Traffic, CDN performance | Free

14. Disaster Recovery (Updated)

Scenario | Recovery Approach | RTO | RPO

Node.js process crash | PM2 auto-restarts | < 30 sec | 0

Oracle VM restart | PM2 startup on boot | < 2 min | 0

Database issue | Neon point-in-time restore (7 days) | < 30 min | < 5 min

R2 storage issue | Cloudflare R2 has 99.999999999% durability | N/A | N/A

Full VM failure | Re-provision Oracle Cloud VM, deploy from GitHub | < 1 hour | < 5 min

End of Technical Requirements Document (Updated — Free Stack)

All services listed are permanently free, not trial-based.